

■ AI Data Analyst - Executive Analysis Report

■ Executive Data Analysis Report

1. Executive Summary

- **Primary category performance:** All eight product categories generate $\approx \$190\text{--}\216 k in total revenue, with **Butchers** and **Electric Household Essentials** leading. Average unit prices are tightly clustered between **\$21.5 – \$25.0**.
 - **Discount usage:** A discount is applied on **33.55 %** of all transactions, uniformly across categories ($\approx 31\text{--}35$ %).
 - **Monthly trends:** Revenue per month oscillates between **\$38 k** and **\$51 k**. The most recent full year (2023–2024) shows a modest upward trajectory, while **Jan–2025** is an outlier dip ($\approx \$27$ k). Transaction counts stay around **300–390** per month.
 - **Payment methods:** Cash, Digital Wallet, and Credit Card each capture roughly **one–third** of transactions, indicating a well-balanced payment ecosystem.
- > **Bottomline for executives:** The business enjoys stable, diversified sales with consistent discounting and payment-method mix. The sharp dip in Jan–2025 should be investigated (inventory, promotions, or external factors).

2. Dataset Overview & Data Quality Profile

| Attribute | Value |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Duplicate rows** | 0 |

| **Missing values** | *Item*, *Price Per Unit*, *Quantity*, *Total Spent*, *Discount Applied* |

| **Potential date column** | *Transaction Date* (string → datetime) |

| **Data quality notes** | Missing values were imputed; date column converted; whitespace trimmed. |

3. Data Cleaning & Validation Summary

| Step | Action |

|-----|-----|

- | **1. Duplicate removal** | `drop_duplicates()` (none found) |
- | **2. Whitespace trimming** | Stripped all object columns |
- | **3. Date conversion** | `Transaction Date` → `datetime` (coerce errors) |
- | **4. Numeric missing values** | Filled with column median |
- | **5. Boolean handling** | Normalised `Discount Applied` to `bool`, missing → `False` |
- | **6. Remaining object columns** | Replaced literal `'nan'` with `np.nan`, then filled with `'Unknown'` |
- | **Validation** | All rows retained (12,575). No integrity violations after cleaning. |

The cleaned DataFrame is stored in the variable `result`.

4. User Query & Analytical Approach

User question

> “Generate a complete PowerBI-style executive dashboard overview with primary category breakdown and trends.”

Analytical methodology

1. **Category-level summary** – `groupby('Category')` to compute:
 - Total revenue (`sum Total Spent`)
 - Total quantity (`sum Quantity`)
 - Average price per unit (`mean Price Per Unit`)
 - Number of distinct transactions (`nunique Transaction ID`)
 - Discount usage rate (`mean Discount Applied`)
2. **Monthly revenue & transaction trend** –
 - Create a `Month` period (`dt.to_period('M')`)
 - Aggregate revenue and distinct transaction count per month
3. **Payment-method distribution** – `value_counts()` on `Payment Method`.
4. **Overall discount rate** – overall mean of `Discount Applied`.

All results are converted to JSON-serialisable structures for downstream visualisation.

5. Python Analysis Code & Executed Data Output

5.1 Code

```
# Ensure the date column is datetime df['Transaction Date'] = pd.to_datetime(df['Transaction Date']) # 1. Category level summary category_summary = ( df.groupby('Category') .agg( total_revenue=('Total Spent', 'sum'), total_quantity=('Quantity', 'sum'), avg_price_per_unit=('Price Per Unit', 'mean'), transaction_count=('Transaction ID', 'nunique'), discount_rate=('Discount Applied', 'mean') ) .reset_index() ) # 2. Monthly revenue & transaction trend monthly_trend = ( df.assign(Month=df['Transaction Date'].dt.to_period('M')) .groupby('Month') .agg( monthly_revenue=('Total Spent', 'sum'), monthly_transactions=('Transaction ID', 'nunique') ) .reset_index() .astype({'Month': 'str'}) ) # 3. Payment method distribution payment_method_distribution = ( df['Payment Method'] .value_counts() .reset_index() .rename(columns={'index': 'Payment Method', 'Payment Method': 'Count'}) ) # 4. Overall discount usage rate overall_discount_rate = df['Discount Applied'].mean() # Package results result = { "category_summary": category_summary.to_dict(orient="records"), "monthly_trend": monthly_trend.to_dict(orient="records"), "payment_method_distribution": payment_method_distribution.to_dict(orient="records"), "overall_discount_rate": float(overall_discount_rate) }
```

5.2 Executed Output (JSON style)

```
{ "category_summary": [ {"Category": "Beverages", "total_revenue": 204751.0, "total_quantity": 8784.0, "avg_price_per_unit": 23.300574345883856, "transaction_count": 1567, "discount_rate": 0.3324824505424378}, {"Category": "Butchers", "total_revenue": 215930.0, "total_quantity": 8638.0, "avg_price_per_unit": 25.013711734693878, "transaction_count": 1568, "discount_rate": 0.34630102040816324}, {"Category": "Computers and electric accessories", "total_revenue": 199481.0, "total_quantity": 8758.0, "avg_price_per_unit": 23.146341463414632, "transaction_count": 1558, "discount_rate": 0.3433889602053915}, {"Category": "Electric household essentials", "total_revenue": 211951.0, "total_quantity": 8759.0, "avg_price_per_unit": 24.317096165933375, "transaction_count": 1591, "discount_rate": 0.32809553739786296}, {"Category": "Food", "total_revenue": 203600.5, "total_quantity": 8873.0, "avg_price_per_unit": 23.088790931989923, "transaction_count": 1588, "discount_rate": 0.3198992443324937}, {"Category": "Furniture", "total_revenue": 202471.0, "total_quantity": 8858.0, "avg_price_per_unit": 23.348837209302324, "transaction_count": 1591, "discount_rate": 0.33626649905719674}, {"Category": "Milk Products", "total_revenue": 187815.5, "total_quantity": 8765.0, "avg_price_per_unit": 21.505681818181817, "transaction_count": 1584, "discount_rate": 0.3409090909090909}, {"Category": "Patisserie", "total_revenue": 191605.0, "total_quantity": 8465.0, "avg_price_per_unit": 23.06380890052356, "transaction_count": 1528, "discount_rate": 0.3370418848167539} ], "monthly_trend": [ {"Month": "2022-01", "monthly_revenue": 55298.5, "monthly_transactions": 390}, {"Month": "2022-02", "monthly_revenue": 45278.5, "monthly_transactions": 331}, {"Month": "2022-03", "monthly_revenue": 42515.0, "monthly_transactions": 353}, {"Month": "2022-04", "monthly_revenue": 41744.0, "monthly_transactions": 346}, {"Month": "2022-05", "monthly_revenue": 41975.0, "monthly_transactions": 330}, {"Month": "2022-06", "monthly_revenue": 44420.5, "monthly_transactions": 338}, {"Month": "2022-07", "monthly_revenue": 46858.5, "monthly_transactions": 371}, {"Month": "2022-08", "monthly_revenue": 43069.5, "monthly_transactions": 347}, {"Month": "2022-09", "monthly_revenue": 47307.0, "monthly_transactions": 348}, {"Month": "2022-10", "monthly_revenue": 39874.0, "monthly_transactions": 317}, {"Month": "2022-11", "monthly_revenue": 44077.5, "monthly_transactions": 354}, {"Month": "2022-12", "monthly_revenue": 40588.0, "monthly_transactions": 309}, {"Month": "2023-01", "monthly_revenue": 49246.0, "monthly_transactions": 373}, {"Month": "2023-02", "monthly_revenue": 40842.0, "monthly_transactions": 314}, {"Month": "2023-03", "monthly_revenue": 40813.0, "monthly_transactions": 311}, {"Month": "2023-04", "monthly_revenue": 40533.0, "monthly_transactions": 323}, {"Month": "2023-05", "monthly_revenue": 42216.5, "monthly_transactions": 334}, {"Month": "2023-06", "monthly_revenue": 43884.5, "monthly_transactions": 349}, {"Month": "2023-07", "monthly_revenue": 47694.0, "monthly_transactions": 375}, {"Month": "2023-08", "monthly_revenue": 39677.0, "monthly_transactions": 317}, {"Month": "2023-09", "monthly_revenue": 42588.0, "monthly_transactions": 333}, {"Month": "2023-10", "monthly_revenue": 39949.5, "monthly_transactions": 308},
```

```
{
  "Month": "2023-11", "monthly_revenue": 38316.0, "monthly_transactions": 302,
  "Month": "2023-12", "monthly_revenue": 44974.0, "monthly_transactions": 348,
  "Month": "2024-01", "monthly_revenue": 50295.5, "monthly_transactions": 385,
  "Month": "2024-02", "monthly_revenue": 38881.0, "monthly_transactions": 320,
  "Month": "2024-03", "monthly_revenue": 44923.0, "monthly_transactions": 355,
  "Month": "2024-04", "monthly_revenue": 48007.0, "monthly_transactions": 327,
  "Month": "2024-05", "monthly_revenue": 46370.5, "monthly_transactions": 369,
  "Month": "2024-06", "monthly_revenue": 46565.5, "monthly_transactions": 351,
  "Month": "2024-07", "monthly_revenue": 42381.5, "monthly_transactions": 343,
  "Month": "2024-08", "monthly_revenue": 45640.5, "monthly_transactions": 375,
  "Month": "2024-09", "monthly_revenue": 43463.5, "monthly_transactions": 340,
  "Month": "2024-10", "monthly_revenue": 44364.0, "monthly_transactions": 353,
  "Month": "2024-11", "monthly_revenue": 45703.5, "monthly_transactions": 347,
  "Month": "2024-12", "monthly_revenue": 50528.0, "monthly_transactions": 376,
  "Month": "2025-01", "monthly_revenue": 26742.0, "monthly_transactions": 213 } ],
  "payment_method_distribution": [ { "Payment Method": "Cash", "Count": 4310 }, { "Payment Method": "Digital Wallet", "Count": 4144 }, { "Payment Method": "Credit Card", "Count": 4121 } ],
  "overall_discount_rate": 0.33550695825049703 }
}
```

6. Key Insights & Strategic Recommendations

■ Insights

| Insight | Detail |

|-----|-----|

| **Revenue leaders** | *Butchers* (\$215.9 k) and *Electric Household Essentials* (\$212.0 k) are the top grossing categories. |

| **Uniform discounting** | Discount applied in $\approx 33\%$ of all sales, with little variation across categories (31–35%). |

| **Stable monthly performance** | Revenue stays within a \$38–\$51 k band; transaction count stays near 350 ± 40 per month. |

| **Payment channel parity** | Cash, Digital Wallet, and Credit Card each capture $\sim 1/3$ of transactions – no channel is over- or under-utilised. |

| **Anomalous dip** | Jan 2025 shows a sharp revenue/transaction drop ($\approx \$27$ k, 213 transactions). Could be seasonal, inventory, or data capture issue. |

■ Recommendations

1. Investigate Jan 2025 dip

- Cross-check inventory levels, promotional calendar, and external events (e.g., holidays, supply chain disruptions).
- If it's a data anomaly, correct the source; if it's a genuine slowdown, consider targeted promotions.

2. Leverage high